

SPIKE Library Installation Instructions

1. Install MATLAB

- a. Install MATLAB R2024b or later by following the MathWorks installation guide:
<https://www.mathworks.com/help/install/install-products.html>

The following products are required:

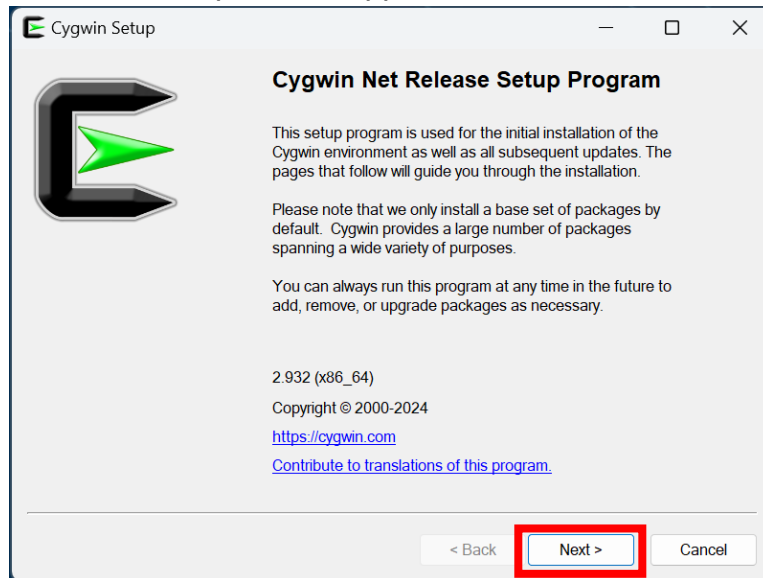
- MATLAB
 - Simulink
 - MATLAB Coder
 - Simulink Coder
 - Embedded Coder
- b. After launching MATLAB, enter 'ver' in the Command Window to verify that all required products are installed.

Command Window

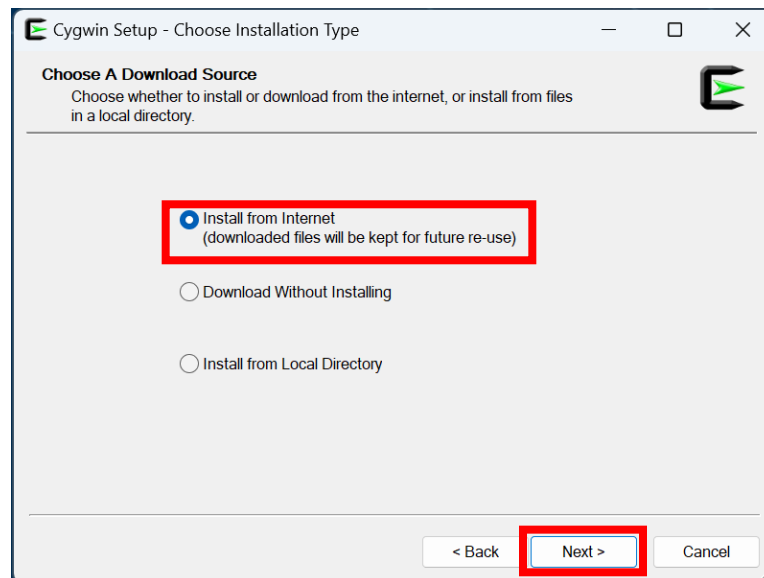
fx >> ver

2. Install Cygwin

- a. Download the Cygwin installer from the official website: <http://www.cygwin.com/>
- b. Launch setup.exe after downloading.
- c. When the setup screen appears, click '**Next**'.

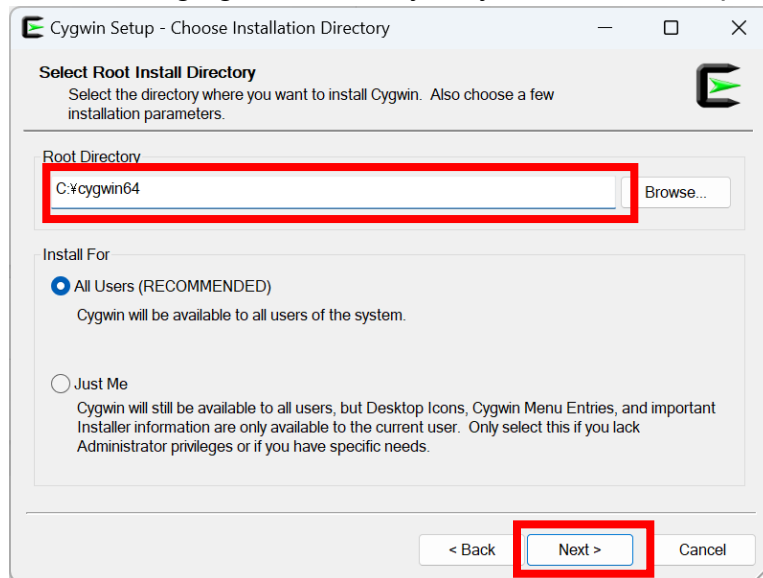


- d. Select '**Install from Internet**' and click '**Next**'.



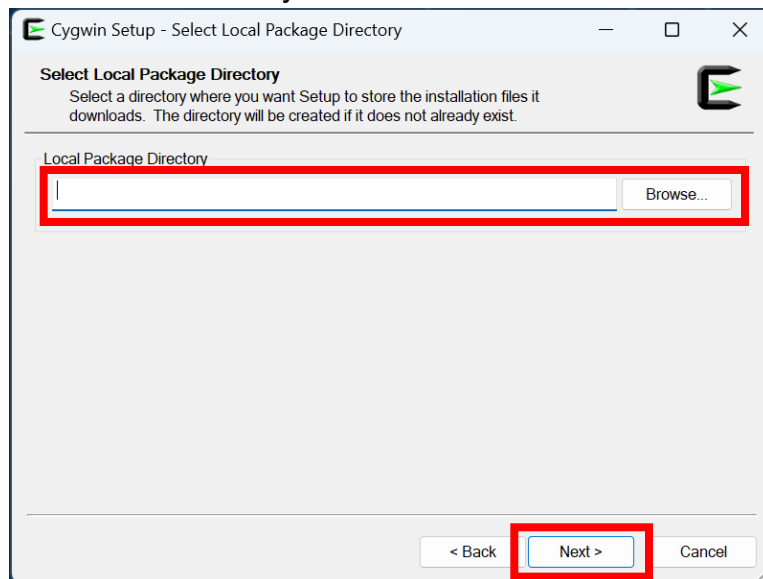
- e. Set the installation directory to C:\cygwin64 and click '**Next**'.

Note: Changing this directory may affect the build process.

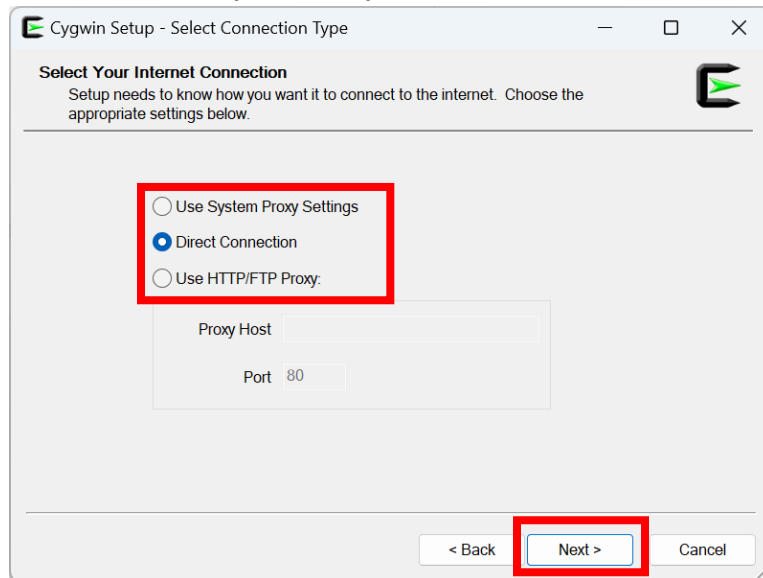


- f. Select a local directory where the installation packages will be downloaded, then click '**Next**'.

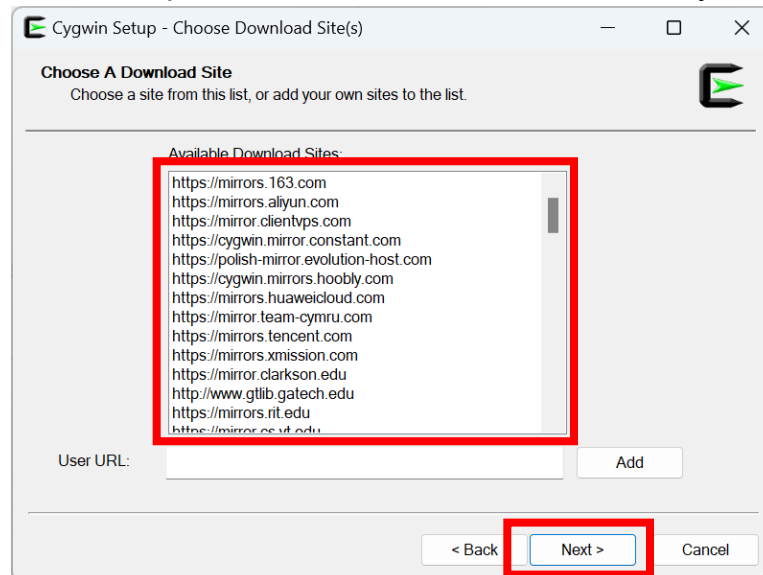
Note: This directory can be deleted after the installation is complete.



- g. Select the appropriate Internet connection method for your environment.
If there are no special requirements, select '**Direct Connection**' and click '**Next**'.



- h. Select a download server. Any server may be used; however, choosing one geographically close to your location is recommended for better performance. Servers using https may be more stable than those using ftp. If the download is extremely slow or interrupted, select a different server and retry.



- i. When the Cygwin package list appears, set '**View**' to '**Full**' (top left of the window). Select the following packages for installation (only the versions shown have been tested):

- gcc-core

View

Full

 Search

gcc-core

Clear

☐ Keep

☒ Best

☐ Sync

☐ Test

Package	Current	New	Src?	Categories
diagn-gcc-core		Skip	☐	Devel, Unmaintained
gcc-core		12.4.0-3	☐	Devel
mingw64-i686-gcc-core		Skip	☐	Devel
mingw64-x86_64-gcc-core		Skip	☐	Devel

- make

View

Full

 Search

make

Clear

☐ Keep

☒ Best

☐ Sync

☐ Test

Package	Current	New	Src?	Categories
make		4.4.1-2	☐	Devel

- ruby

View Full Search ruby Clear ☐ Keep ☒ Best ☐ Sync ☐ Test

Package	Current	New	Src?	Categories
ruby		2.6.4-1	☐	Interpreters, Ruby

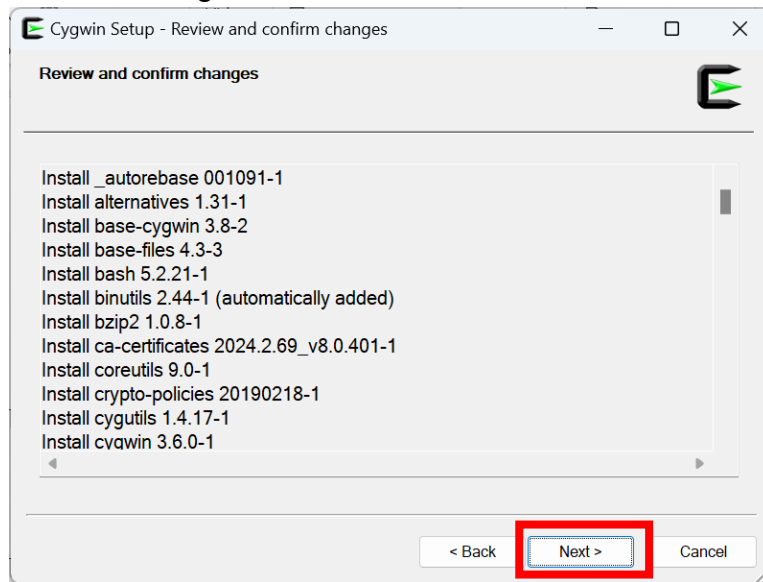
- rubygems

View Full Search rubygems Clear ☐ Keep ☒ Best ☐ Sync ☐ Test

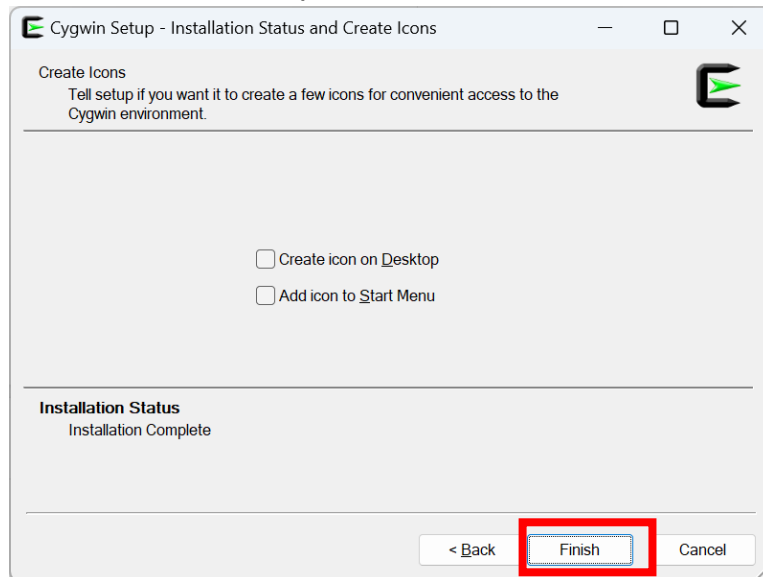
Package	Current	New	Src?	Categories
rubygems		3.5.11-1	☐	Ruby

After selecting the packages, click '**Next**'.

j. Click **'Next'** again to start the installation.



k. Click **'Finish'** to complete the installation.



3. Install Arm GNU Toolchain

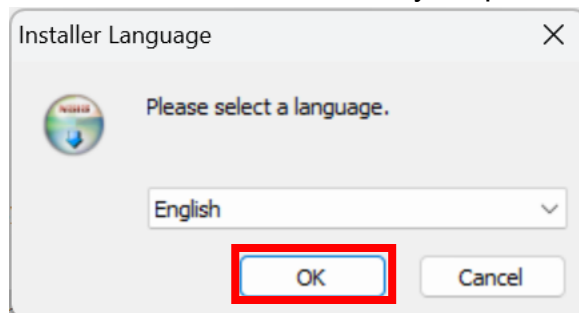
- a. Download the Arm GNU Toolchain installer from the Arm Developer website:
<https://developer.arm.com/downloads/-/gnu-rm>

The recommended version (tested) is: gcc-arm-none-eabi-10.3-2021.10-win32.exe.

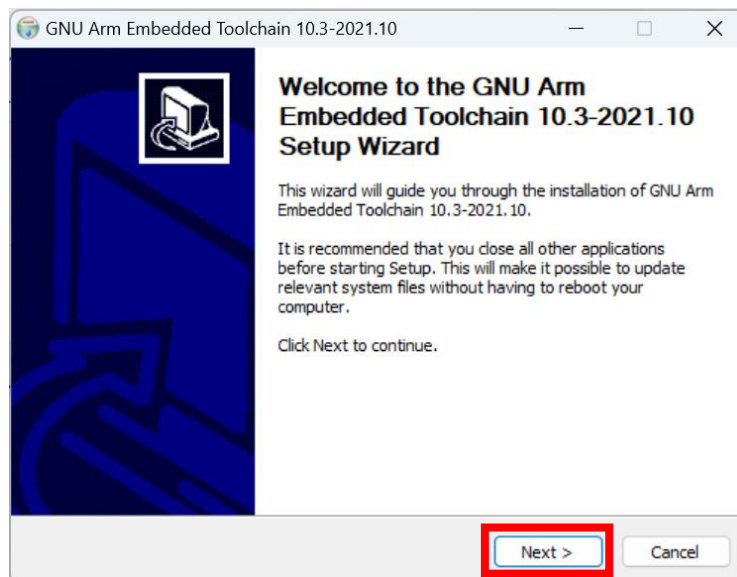
The latest version of the installer is available below. However, we have not tested its functionality.

<https://developer.arm.com/Tools%20and%20Software/GNU%20Toolchain>

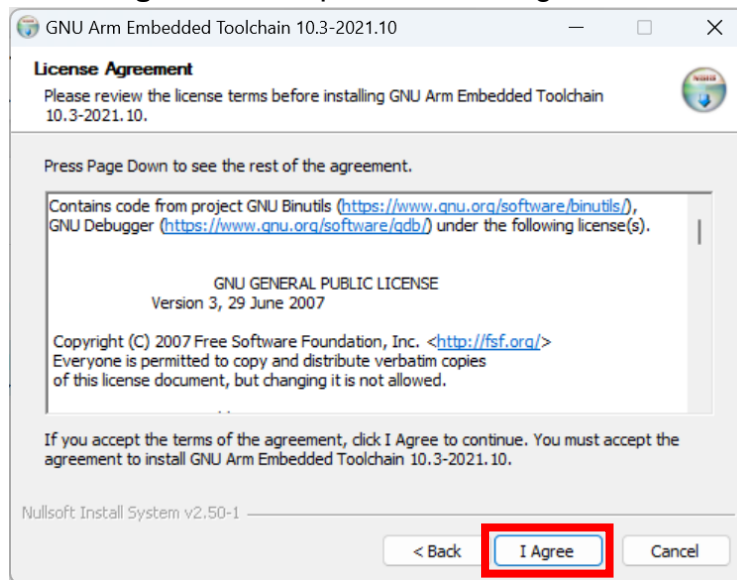
- b. Start the installer and select your preferred language.



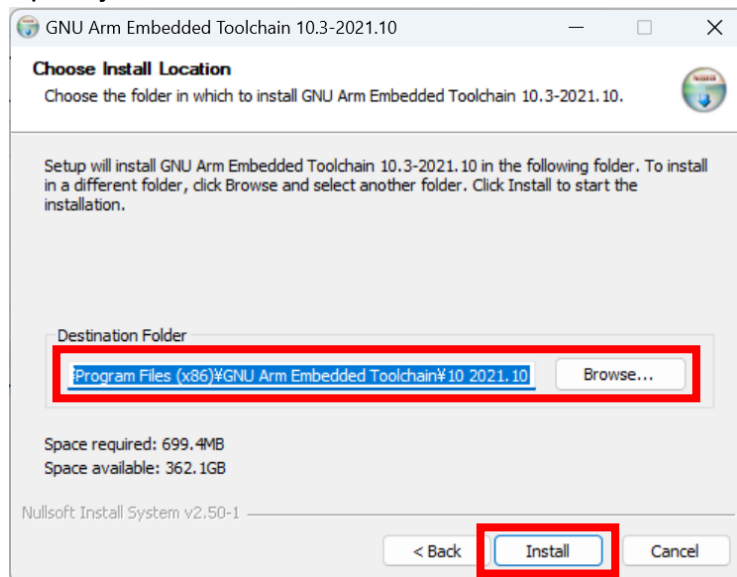
- c. Click 'Next'.



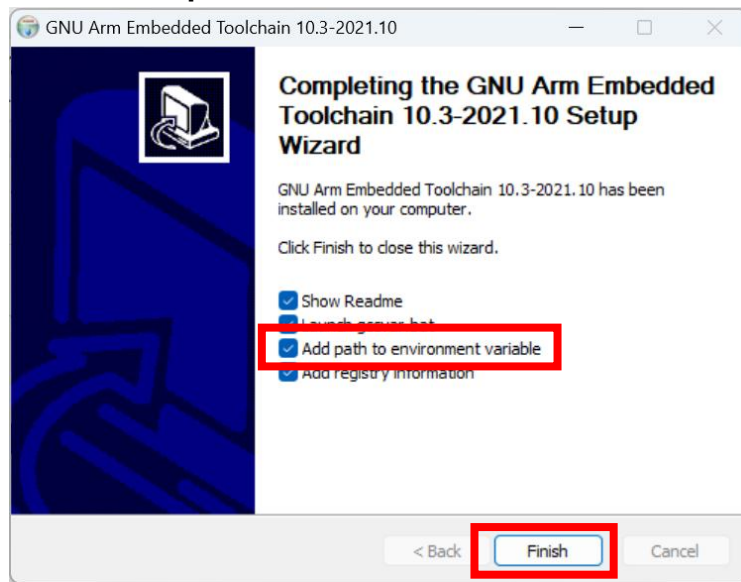
- d. Click '**I Agree**' to accept the license agreement.



- e. Specify the installation folder and click '**Install**'.

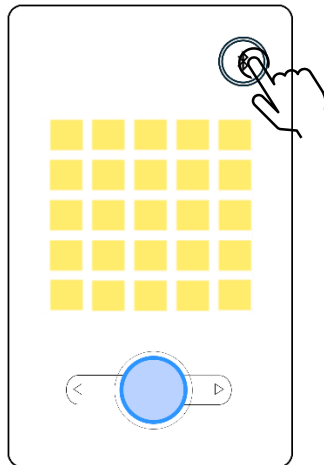


- f. Check **'Add path to environment variable'**, then click **'Finish'**.

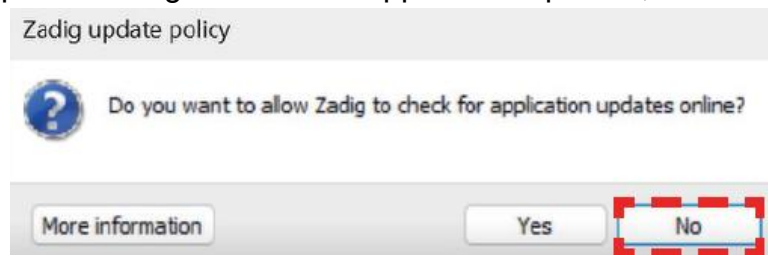


4. Install USB driver

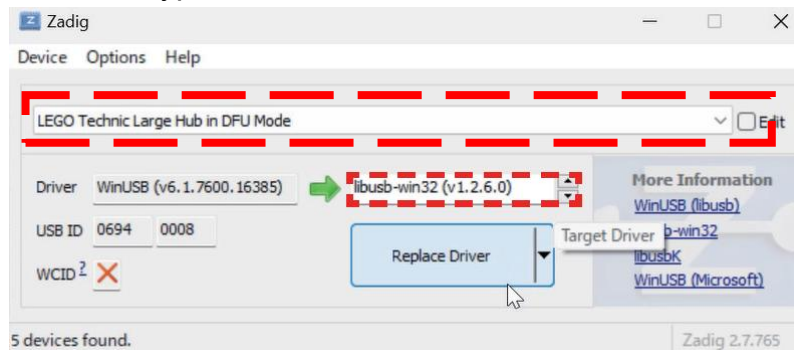
- a. Download the Zadig USB driver from the official website: <https://zadig.akeo.ie/>
The tested version is zadig-2.7.exe
- b. Turn off the power of the SPIKE Prime Large Hub.
While holding down the Bluetooth button on the SPIKE, connect it to the computer using a USB cable.
When connected, the Bluetooth button will light up in purple. Continue holding the button until it repeatedly cycles through the following sequence: red → green → blue → off → red →
Then release the Bluetooth button.



- c. Launch the downloaded Zadig executable file.
If a dialog appears asking to check for application updates, click '**No**'.



- d. The Zadig main window will open. Ensure that '**LEGO Technic Large Hub in DFU Mode**' is selected in the device list near the center of the window. Next, set the driver type to '**libusb-win32**', then click '**Install WCID Driver**'.



[Note] About selecting 'LEGO Technic Large Hub in DFU Mode'

If LEGO Technic Large Hub in DFU Mode does not appear in the device list:

- Confirm that the SPIKE is properly connected to the computer.
- Ensure that the Bluetooth button on the SPIKE is flashing in rainbow colors.
- Alternatively, select '**Options**' → '**List All Devices**' from the menu.

